

**HINTS FOR PAINTERS,
DECORATORS,
AND PAPER-HANGERS.**

BEING

**A SELECTION OF USEFUL RULES, DATA, MEMORANDA, METHODS,
AND SUGGESTIONS FOR HOUSE, SHIP AND FURNITURE PAINTING,
PAPER-HANGING, GILDING, COLOR MIXING, AND OTHER
MATTERS USEFUL AND INSTRUCTIVE TO PAINTERS
AND DECORATORS.**

Prepared with Special Reference to the Wants of Amateurs.

BY

AN OLD HAND.

**NEW YORK:
THE INDUSTRIAL PUBLICATION COMPANY.
1882.**

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PREFACE.

This little work is intended to furnish the PRACTICAL HOUSE-PAINTER—THE EVERY-DAY WORKMAN—with information sufficient to enable him to understand his business intelligently. One of its objects is to deal with the nature, characteristics, qualities, and defects of the materials employed by the class of artisans for whom it is written; and to a limited extent this has been done with as little theory as possible; high-sounding technicalities have also been avoided wherever the author has been able to make himself understood without them.

It is thought the young painter may derive great profit and advancement from a careful study of this book, as the hints, rules, and recipes it contains are reliable, practical, and of every-day use.

The author has consulted many works on the subject, and is indebted to many of them for much of the matter contained, among which may be mentioned “Building Construction and Materials,” “House-Painter’s Hand-Book,” “Artists’ and Tradesmen’s Companion,” “Painter’s Guide,” “Chevreul’s Oils and Paints,” and several other works of more or less note. To this has been added many things discovered by the actual experience of the writer.

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HINTS FOR PAINTERS AND PAPER-HANGERS.

There is a general belief abroad that anybody can execute all that is required of a house-painter. This is a very popular error; it is not so easy to prepare and apply a coat of paint in a thorough workmanlike manner as some may imagine. It is still less easy to paint in parti colors; and very few can produce a good piece of graining.

The painter should not only be acquainted with the method of applying paint when it is provided for him, and the brush placed in his hand, but he should know the composition of the colors; the manner in which they are made, and the colors which must harmonize with each other when they are associated together. These observations being of a practical nature and the result of experience, are commended for his perusal and study.

Preparation.—All surfaces painted should be first thoroughly dry and free from dust. All heads of nails should be punched or “set” below the surface of the wood, and after the priming is dry, the holes formed by the heads, as well as all cracks, defects, etc., should be filled up with putty made of raw linseed oil and whiting. To prevent knots or “fat” spots from showing through, they should receive two coats of shellac varnish, and when dry rubbed down with sand-paper. The shellac should be applied with a small brush. It dries soon and may be painted over as soon as dry. Shellac prepared this way is called “knotting,” and can be purchased wherever painters’ materials are kept for sale.

Materials.—The principal materials used for painting, *i.e.*, white lead and oxide of zinc, are so well known that it is unnecessary to allude particularly to their manufacture at present. Before using them they should be mixed with pure raw linseed oil. Turpentine may be used in cold weather to make the paint work easy, as the oil is apt to chill, which thickens it and makes it difficult to apply. In warm weather, however, turpentine should not be used in priming any parts where the sun shines upon. In cold weather it is always necessary to use litharge or some other drier; or the work will remain a long time before it hardens; in summer, however, driers in most cases are unnecessary, for if the wood to be painted is as well seasoned as it ought to be there is little danger of the paint being washed off by rain, as it will mostly be absorbed in the surface.

To make satisfactory work it is imperative that all cans, pots and brushes used in painting be perfectly clean at the start, and kept so whenever the conditions will permit. A marble slab and muller will be required to grind the finer colors used. Sometimes a small cast-iron mill will be found useful not only to grind colors; but to pass the tinted color through, so that it may be thoroughly mixed. It is scarcely necessary to say that it is presumed the workman will know what brushes he will require according to the work he has in hand. His large ground paint brush, called a “pound-brush;” his half-size, for smaller work; his fitch and sash tools, duster, pallet, putty and hasp knives; oval and flat varnish brushes, varnish-pot, step-ladders and long ladders, mahl-stick, pallet-board, gilding-knife, camel and sable hair-pencils, whitewash brushes, jack for window work, cushion, tip-pole, etc., etc.

Large brushes, such as 6-0 size, should be bridled when new by winding good twine around them about one-third up the length of the bristles, and as the brush wears, this binding can be unwound. Care should be taken to use the brushes so that they will have a flat wedge-shaped point, straight on the edge. This can be done by holding the brush always in one position. If brushes are turned round and round in the hand while in use, they wear round and stubby on the point and soon become useless for fine or smooth work.

Brushes that are in use every day should be placed in water half the depth of the bristles at night, taking care that brushes containing different colors do not come in contact with one another. If they are to be laid aside for any length of time, however, they should be washed with warm water and soap after being thoroughly cleansed with turpentine, and laid away in a moist place.

As a general thing it is better to buy putty already made at a regular paint store, where you may depend upon its being made of good whiting and linseed oil than to make it. Putty should not be used until the work has been primed, for new paint holds the putty very firmly.

White lead is to be judged of by being well ground and possessing the mellowness given to it by age.

It is well known amongst painters that the best article is the most economical, as it works out with more ease, and repays the difference of cost in its better appearance and extra durability. Linseed oil is also better for having due age, for the same reasons as the white lead, working with softness and advantage after parting with the water, which is generally combined with new oil.

In most cases *driers* are added to paints to cause them to dry more quickly, and a *solvent* is sometimes required to make the paints work more freely. When the color required differs from that of the main paint used, the desired tint is obtained by adding a staining or coloring pigment. The materials generally employed may, for convenience, be classed as follows:

Bases.—White lead, red lead, zinc white, oxide of iron. **Vehicles.**—Oils, spirits of turpentine. **Solvents.**—Spirits of turpentine. **Driers.**—Litharge, acetate of lead, sulphate of zinc and binoxide of manganese, red lead, etc. **Coloring Pigments.**—Ochres, lampblack, umber, sienna, and many metallic salts that will be hereinafter mentioned.

White lead may be obtained either pure or mixed with various substances, such as sulphate of baryta, sulphate of lead, whiting, chalk, zinc white, etc. These substances do not combine with oil as well as does white lead, nor do they so well protect any surface to which they are applied. Sulphate of baryta, the most common adulterant, is a dense, heavy, white substance, very like white lead in appearance. It absorbs very little oil, and may frequently be detected by the gritty feeling it produces when the paint is rubbed between the finger and thumb.

White Lead is sold either dry, in powder or lump, or else ground in oil in a paste containing from 7 to 9 per cent. of linseed oil, and more or less adulterated, unless specially marked "genuine." When slightly adulterated with a very white sulphate of baryta, like that of the Tyrol, the mixture is considered preferable for certain kinds of work, as the barytes communicates opacity to the color and protects the lead from being speedily darkened by sulphurous smoke or vapors. White lead improves by keeping, and when of good quality, will go much further and last much better than when employed fresh; moreover, paint made with

new lead has a tendency to become yellow. It should not be exposed to the air or it will turn grey.

Of all the bases used for paints, white lead is the most commonly used, and for surfaces of wood it affords in most cases the best protection, being dense, of good body, and permanent. It has the disadvantage, however, of blazening when exposed to sulphur acids, and of being injurious to those who handle it.

Red lead is produced by raising *massicot* (the commercial name for oxide of lead) to a high temperature, short of fusion, during which it absorbs oxygen from the air and is converted into red lead or *minium*, an oxide of lead. The color is lasting, and is unaffected by light when it is pure and used alone, but any preparation containing lead or acids mixed with it deprive it of color, and impure air makes it black. It may be used for a drier, as it possesses many of the properties of litharge; it is also often employed in painting wrought iron work, to which it adheres with a tenacity not equalled by any other paints; it is sometimes objected to for this purpose, on the ground that galvanic action is set up between the lead and the iron. It is also frequently used for priming on wood work, and is especially adapted for hard woods. It is frequently adulterated with brick dust; this may be detected by heating the powder in a crucible, and treating it with dilute nitric acid; the lead will be dissolved but the brick dust will remain. It is also adulterated with colcother, a sesquioxide of iron. Sulphide of antimony, or antimony vermilion, is sometimes used as a substitute for red lead. It is sold in a very fine powder, without taste or smell, and which is insoluble in water, alcohol, or essential oils. It is but little acted upon by acids, and is said to be unaffected by air or light. It is adapted for mixing with white lead, and affords an intensely bright color when ground in oil.

Oxide of zinc, or zinc white, is durable in water or oil; it dissolves in hydrochloric acid; it does not blacken in the presence of sulphuretted hydrogen; and it is not injurious to the men who make it, or to the painters who use it; but on the other hand, it does not combine with oil well, and is wanting in body and covering power, and is difficult to work. It is easily acted upon by the carbonic acid in rain water, which dissolves the oxide, and it therefore is unfit for outside work. The acids contained in unseasoned wood also have a great effect upon it. When pure and used for inside work, it retains its color well, and will stand washing for many years without losing any of its freshness. When dry it becomes very hard, and will take a fine polish. This paint is suitable for any place that is subjected to vapors containing sulphur, or in places where foul air is emanated from decaying animal matter. In such positions, of course, zinc paints should not be mixed with “patent” or other driers which contain lead. The best driers to employ with it are sulphate of manganese and sulphate of zinc. This white is recommended as being preferable to white lead for painting on a dark ground. The reason of this is that the soap formed by the combination of the lead and oil in lead paints is semi-transparent, and the dark ground shows through it. The want of density, however, in zinc paints, is a great drawback to their use, and the purest zinc oxide is not always the best for paint on account of its low specific gravity; and in this respect, the American zinc whites, which are frequently very pure, do not generally give as good satisfaction as the zinc whites made in Belgium.

Vehicles.—Oils are divided into two classes—fixed oils and volatile oils. Fixed oils are extracted by pressure from vegetable substances, they are of a fatty nature, do not evaporate in drying, and will bear a temperature short of 500° Fahr., without decomposing. They are subdivided into drying oils, which become thick upon exposure to air. Of these, linseed oil is

most commonly used as an ingredient for paint. Its qualities when pure are excellent, and it may be considered the best of all oils for use in paint, putty, and other similar substances. It oxidizes and becomes thick upon exposure to the air. This property is very much increased by adding other substances to it and boiling them together. It is superior in drying powers, tenacity, and body to any other fixed oil. The best oil comes from the Black Sea and the Baltic; that from the East Indian seed is inferior, as the seed is less carefully cleaned, and contains too much stearine. Raw linseed oil is clear and light in color, works smoothly, and is used for internal works, for delicate tints, and for grinding up colors. Boiled oil is much thicker, darker, and more apt to clog. It is used for outside work, as its greater body and rapidity in drying make it a quicker and more efficient protection.

Volatile Oils are generally obtained by distillation, and have an odor resembling that of the plant from which they are obtained. They are, as a rule, colorless at first, but upon exposure to air and light they become darker, thicker, and eventually are converted into a kind of resin. Spirits of turpentine is the best variety of this class for mixing with ordinary paints. Naphtha and benzine are sometimes used instead of turpentine, but not often, and their use is not recommended when the latter can be obtained. Good spirits of turpentine is lighter in weight and more inflammable than bad. It is colorless and has a pleasant pungent smell, whereas the smell of inferior qualities is disagreeable. It is used as a solvent for resins and other substances in making varnishes; also in paint to make it work more smoothly. It is useful also in flattening coats, but will not stand exposure to the weather.

Driers.—Driers are substances added to paint in order to cause the oils to thicken and solidify more rapidly. The action of these substances is not thoroughly understood. Chevreuil has shown that the drying of linseed oil is caused by the absorption of oxygen; and there can be no doubt that for the most part driers act as carriers of oxygen to the oil, a very small quantity producing considerable effects.

The best driers are those which contain a large proportion of oxygen, such as litharge, acetate of lead, red lead, sulphate of zinc, verdigris, etc. They are sometimes used to improve the drying qualities of the oil with which the paint is mixed, or they may themselves be ground up with a small quantity of oil, and added to the paint just before it is used.

Litharge or oxide of lead is the drier most commonly used, and is produced in extracting lead from its ores. It can be produced on a small scale by scraping off the dross which forms on molten lead exposed to a current of air. *Massicot* is a superior kind of litharge, being produced by heating lead to an extent insufficient to fuse the oxide. Sugar of lead, or, as it is more frequently called, acetate of lead, ground in oil, and copperas and white vitriol (sulphate of zinc), are also used as driers, especially for light tints. Oxide of manganese is quicker in its effects, but is of a very dark color, and seldom used except for deep tints. Japanners' gold size and verdigris (acetate of copper) are also much used for dark colors. Care must be taken not to apply too much of the size, or it will make the paint brittle. Red lead (oxide of lead) is often used as a drier when its color will not interfere with the tint required. It is not so rapid in its action as litharge or massicot. Sulphate of manganese is the best drier for zinc white, about 6 or 8 ounces only being used for 100 lbs., of ground zinc white paint. The manganese should be mixed with a small quantity of the paint first, and then added to the bulk. If great care be not taken in mixing the drier the work will be spotted. Sulphate of zinc is also a good drier for zinc paint.

Patent driers contain oxidizing agents, such as litharge or acetate of lead ground and mixed in oil, and therefore in a convenient form for immediate use. There is great danger, however, in using such driers, unless they are of the best quality from a reliable maker. Some of the inferior descriptions depend for their drying qualities upon lime.

The following points should be observed in using driers:—

1st. Not to use them unnecessarily with pigments which dry well in oil color.

2d. Not to employ them in excess, which would only retard the drying.

3d. Not to add them to the color until about to be used.

4th. Not to use more than one drier to the same color.

5th. To avoid the use of patent driers, unless known to be of good quality.

6th. To avoid the use of driers in the finishing coat of light colors, as they are liable to injure the color.

Coloring Pigments.—It will be impossible in a small work of this sort to give anything like a complete list of the pigments used to produce the colors and tints used by the house painter and decorator. A few of the most useful may, however, be mentioned. It is not proposed to give a detailed description of them, but merely to distinguish those that are injurious from the others. Many of these, such as the ochres, umbers, etc., are from natural earths; others are artificially made. They may generally be purchased either in the form of dry powder or ground in oil.

Blacks.—Lampblack is the soot produced by burning oil, resin, small coal, resinous woods, coal tar or tallow. It is in the state of very fine powder; works smoothly; is of a dense black color and durable, but dries very slowly in oil.

Vegetable black is a better kind of lampblack made from oil. It is very light, free from grit, and of a good color. It should be used with boiled oil, driers, and a little varnish. Raw linseed oil or spirits of turpentine keeps it from drying.

Ivory-black is obtained by calcining waste ivory in close vessels and then grinding. It is intensely black when properly burned. Bone-black is inferior to ivory-black, and prepared in a similar manner from bones. Blue-black and Frankfort black of the best quality are made from vine twigs; inferior qualities from other woods charred and reduced to powder. In Europe some other blacks are used, but we seldom meet with them in this country.

Blues.—Prussian blue is made by mixing prussiate of potash with a salt of iron. The prussiate of potash is obtained by calcining and digesting old leather, blood, hoofs, or other animal matter with carbonate of potash and iron filings. This color is much used, especially for dark blues, making purples, and intensifying black. It dries well with oil. Slight differences in the manufacture cause considerable variation in tint and color, which leads to the material being known by different names—such as Antwerp blue, Berlin blue, Hærlem blue, Chinese blue, etc. Indigo is produced by steeping certain plants in water, and allowing them to ferment. It is a transparent color; works well in oil or water, but is not durable, especially when mixed with white lead.

Ultramarine was originally made by grinding the valuable mineral *Lapis lazuli*. Genuine ultramarine so made is very expensive, but artificial French and German ultramarines are made of better color, and cheaply, by fusing and washing and reheating a mixture of soda, silica, alum, and sulphur. This blue is chiefly used for coloring wall papers.

Cobalt blue is an oxide of cobalt made by roasting cobalt ore. It makes a beautiful color, and works well in water or oil.

Smalt, Saxon blue and Royal blue are colored by oxides of cobalt.

There are a few other blues, such as Celestial or Brunswick blue, damp blue and verditer, that are chemical compounds, compounds of alum, copper, lime, and other substances; but they are so seldom used in this country that it is unnecessary to describe them in detail.

Yellows.—Chrome yellows are chromates of lead, produced by mixing dilute solutions of acetate or nitrate of lead and bichromate of potash. This makes a medium tint known as “middle chrome.” The addition of sulphate of lead makes this paler, when it is known as “lemon chrome,” whereas the addition of caustic lime makes it “orange chrome” of a darker color. The chromes mix well with oil and with white lead either in oil or water. They stand the sun well, but like other lead salts, become dark in bad air. Chrome yellow is frequently adulterated with gypsum.

Naples yellow is a salt of lead and antimony, supposed to have been originally made from a natural volcanic product at Naples. It is not so brilliant as chrome, but has the same characteristics. King’s yellow is made from arsenic, and is therefore a dangerous color to handle, or use for internal work. It is not durable, and it injures several other colors when mixed with them. Chinese yellow, arsenic yellow, and yellow orpiment are other names for this yellow.

Yellow ochre is a natural clay colored by oxide of iron, and found abundantly in many parts of the world. It is not very brilliant, but is well suited for distemper work, as it is not affected by light or air. It does not lose its color when mixed with lime washes as many other colors do. There are several varieties of ochres, all having the same characteristics differing only in color which varies from a golden to a dark brown.

Terra De Sienna, or raw Sienna, is a clay, stained with oxides of iron and manganese, and of a dull yellow color. It is durable both in oil and water, and is useful in all work, especially in graining.

Browns.—Browns generally owe their color to oxide of iron. Raw umber is a clay similar to ochre colored by oxide of iron. The best comes from Turkey; it is very durable both in water and in oil; does not injure other colors when mixed with them.

Burnt UMBER is the last mentioned material burnt to give it a darker color. It is useful as a drier, and in mixing with white lead to make a stone color.

Vandyke Brown.—This color is an earthy dark brown mineral; it is durable both in oil and water, and is frequently employed in graining.

Purple Brown is of a reddish-brown color. It should be used with boiled oil—and a little varnish and driers for outside work.

Burnt Sienna is produced by burning raw sienna. It is the best color for shading gold.

Brown Pink is a vegetable color often of a greenish hue. It works well in water and oil, but dries badly, and will not keep its color when mixed with white lead. Spanish brown and brown ochre are clays colored naturally by various oxides.

Reds.—Carmine, made from the cochineal insect, is the most brilliant red color known. It is, however, too expensive for ordinary house painting, and is not durable. It is sometimes used for inside decoration.

Red Lead.—This color has already been described on [page 11](#).

Vermilion.—This is a sulphide of mercury in a natural state as cinnabar. The best comes from China. Artificial vermilion is also made both in China and in this country from a mixture of sulphur and mercury. Genuine vermilion is very durable, but when mixed with red lead, as it is sometimes, it will not stand the weather. It can be tested by heating in a test tube; if genuine it will entirely volatilize. German vermilion is the tersulphide of antimony, and is of an orange-red color.

Indian Red.—This color is a ground hematite ore brought from Bengal; it is sometimes made artificially by calcining sulphate of iron. The tints vary, but a rosy hue is considered the best. It may be used with turpentine and a little varnish to produce a dull surface, drying rapidly, or with boiled oil and a little driers, in which case a glossy surface will be produced, drying more slowly.

Chinese Red and Persian red are chromates of lead, produced by boiling white lead with a solution of bichromate of potash. The tint of Persian red is obtained by the employment of sulphuric acid.

Venetian Red is obtained by heating sulphate of iron produced as a waste product at tin and copper works. It is often adulterated by mixing sulphate of lime with it during the manufacture. When pure, it is often called “bright red.” Special tints of purple and brown are frequently required, which greatly enhance the value of the material. These tints should be obtained in the process of manufacture, and not produced by mixing together a variety of different shades of color. When the tint desired is attempted to be obtained by this latter course it is never so good, and the materials produced are known in the trade as ‘faced colors,’ and are of inferior value.

Rose Pink.—This is made of a sort of chalk or whiting stained with a tincture of Brazil wood. It fades very quickly, but it is used for paperhangings, common distemper, and for staining cheap furniture.

Lakes.—These are made by precipitating colored vegetable tinctures by means of alum and carbonate of potash. The alumina combines with the organic coloring matter and separates it from the solution. The tincture used varies in the different descriptions of lake. The best, made from cochineal or madder, is used for internal work. Drop lake is made by dropping a mixture of Brazil wood through a funnel on to a slab. The drops are dried and mixed into a paste with gum water. It is sometimes called “Brazil wood lake.” Scarlet lake is made from cochineal; so also are Florentine lake, Hamburg lake, Chinese lake, Roman lake, Venetian lake and Carminated lake.

Orange.—Chrome orange is a chromate of lead, brighter than vermilion, but less durable. Orange ochre is a bright yellow ochre burnt to give it warmth of tint; it dries and works well

in water or oil, and is very durable. It is known also as Spanish ochre. Orange red is produced by a further oxidation than is required for red lead. It is a brighter and better color.

Greens.—These, of course, may be made by mixing blue and yellow together, but such mixtures are less durable than those produced direct from copper, arsenic, etc. The latter are, however, objectionable for use in distemper or on wall papers, as they are very injurious to health. Brunswick green of the best kind is made by treating copper with sal-ammoniac. Chalk, lead and alum are sometimes added. It has rather a bluish tinge; dries well in oil, is durable, and not poisonous. Common Brunswick green is made by mixing chromate of lead and Prussian blue with sulphate of baryta. It is not as durable as real Brunswick green. Mineral green is made from bi-basic carbonate of copper; it weathers well. Verdigris is acetate of copper. It furnishes a bluish-green color, durable in oil or varnish, but not in water; it dries rapidly, but requires great care in using owing to its poisonous qualities. Green verditer is a carbonate of copper and lime; is not very durable. Prussian green is made by mixing different substances with Prussian blue. There are a number of other greens made from copper, but they all possess in a greater or lesser degree, the same qualities as the foregoing. Emerald or Paris green is made of verdigris mixed with a solution of arsenious acid. It is of a very brilliant color, but is very poisonous; is difficult to grind, and dries badly in oil. It should be purchased ready ground in oil, as in that case the poisonous particles do not fly about, and the difficulty of grinding is avoided. Scheele's green and Vienna green are also arseniates of copper, and highly poisonous. Chrome green should be made from the oxide of chromium, and is very durable. An inferior chrome green is made, however, by mixing chromate of lead and Prussian blue, as above mentioned, and is called Brunswick green. The chrome should be free from acid or the color will fade; it may be tested by placing it for several days in strong sunlight.

Proportions of Ingredients in Mixed Paint.—The composition of paints should be governed by the nature of the material to be painted. Thus the paints respectively best adapted for painting wood and iron differ considerably. The kind of surface to be covered, *i. e.*, a porous surface requires more oil than one that is impervious. The nature and appearance of the work to be done. Delicate tints require colorless oil; a flatted surface must be painted without oil, which gives gloss to a shining surface. Again, paint used for surfaces intended to be varnished must contain a minimum of oil. The climate and the degree of exposure to which the work will be subjected; thus, for outside work boiled oil is used, because it weathers better than raw oil. Turps is avoided as much as possible, because it evaporates and does not last; if, however, the work is to be exposed to the sun, turps are necessary to prevent the paint from blistering. The skill of the painter also affects the composition; a good workman can lay on even coats with a smaller quantity of oil and turps than a man who is unskilful; extra turps, especially, are often added to save labor. The quality of the materials makes an important difference in the proportions used. Thus more oil and turps will combine with pure than with impure white lead; thick oil must be used in greater quantity than thin oil. When paint is purchased ready ground in oil, a soft paste will require less turps and oil for thinning than a thick paste. Lastly, the different coats of paint vary in their composition; the first coat laid on to new work requires a good deal of oil to soak into the material; on old work the first coat requires turpentine to make it adhere; the intermediate coats contain a proportion of turpentine to make them work smoothly, and to the final coats the coloring materials are added, the remainder of the ingredients being varied as already described, according as the surface is to be glossy or flatted.

The exact proportion of the ingredients best to be used in mixing paints varies according to their quality, the nature of the work required, the climate, and other considerations. The composition of the paint for the different coats also varies considerably. The proportions given in the following table, must, therefore, only be taken as an approximate guide when the materials are of good quality.

Table showing the composition of the different coats of white paint, and the quantities required to cover 100 yards of newly worked pine.

RED WHITE RAW BOILED LEAD. LEAD. LINSEED LINSEED TURPENTINE. DRIERS. REMARKS. OIL. OIL.							
Inside work, 4 coats not flatted.							
	lbs.	lbs.	pts.			lbs.	
Priming	½	16	6	—	—	¼	Sometimes more red lead is used and less drier. *Sometimes just enough red lead is used to give a flesh-colored tint.
2d Coat	*	15	3½	—	1½	¼	
3d Coat	—	13	2½	—	1½	¼	
4th Coat	—	13	2½	—	1½	¼	
Inside work, 4 coats and flatting.							
Priming	1½	16	6	—	½	1-8	
2d Coat	—	12	4	—	1½	1-10	
3d Coat	—	12	4	—	0	1-10	
4th Coat	—	12	4	—	0	1-10	
Flatting	—	9	0	—	3½	1-10	
Outside work, 4 coats not flatted.							
Priming	2	18½	2	2	—	1-8	When the finished color is not to be pure white, it is better to have nearly all the oil boiled oil.
2d Coat	—	15	2	2	½	1-10	

3d Coat	—	15	2	2	$\frac{1}{2}$	1-10	All boiled oil does not work well. For pure white a larger proportion of raw oil is necessary, because boiled oil is too dark.
4th Coat	—	15	3	$2\frac{1}{2}$	0	1-10	

For every 100 square yards, besides the materials enumerated in the foregoing, $2\frac{1}{2}$ lbs. of white lead and 5 lbs. of putty will be required for stopping.

The area which a given quantity of paint will cover depends upon the nature of the surface to which it is applied, the proportion of the ingredients and the state of the weather. When the work is required to dry quickly, more turpentine is added to all the coats.

In repainting old work, two coats are generally required, the old painting being considered as priming. Sometimes another coat may be deemed necessary.

For outside old work exposed to the sun, both coats should contain one pint of turpentine and four pints of boiled oil, the remaining ingredients being as stated in the foregoing table. The extra turpentine is used to prevent blistering.

In cold weather more turpentine should be used to make the paint flow freely.

Operations.—All priming should be rubbed out as far as possible, for if it is flowed out loosely or laid on thick it will be apt to blister and run. In priming over spots, however, where *patching* is being done, it may sometimes be necessary to leave the coat thick or heavy, for new work can never be re-touched and look well, after the work is second coated or finished; for such re-touching would show and spoil the whole work. In priming the paint should flow easily, and the brush should be pressed on to the wood so that the paint will be forced into the pores. In all cases it is a great saving of time to cover as large a surface as convenient before smoothing or finishing off. All work on the same surface should be finished at the one application, if possible, for “laps”—which form the junctions of work done at different times—should be avoided wherever they can, as they are sure to disfigure the work where they exist. Of course, it is sometimes impossible to avoid laps, but where they must occur, care should be taken to make the connection with as little “lap” as possible or it will be certain to show through the work when finished. The defect, if occurring during the second coating, will show much worse than in the priming coat.

In painting, like everything else, a system must be followed to make much headway. Experience, perhaps, is the only effective teacher in this matter; and any rules laid down by us will have but little effect, if the operator does not, or can not systematize. The workman who follows a system will do nearly twice as much work with less labor, than the man who works only by the rule of thumb. In painting blinds, lattices, railings, cut brackets or other similar

work, some method of operation should be adopted. A little observation on the part of the operator, will soon teach him the best methods to adopt in doing any particular kind of work.

Preparing the Work.—In preparing work for painting, too much care cannot be exercised, as succeeding coats and the final result depend very much on the proper condition of the work when the priming coat is applied. First, all the rough places in the wood should be rubbed down with a block covered with sandpaper; and the mouldings and beads should be well cleaned out with sandpaper. Then (and this is a matter of prime importance) every knot, however small, every indication of sap on the wood, or discoloration of any kind, and every appearance of pitch or gum, should be carefully varnished over with white shellac varnish, if the work is to be finished in white or light tints—or with varnish made from unbleached or common shellac, if the work is to be finished in dark shades. The common shellac, in the latter case, answers equally well with the bleached article, and at less cost. This should not, under any circumstances, be neglected, as it is impossible, in the nature of things, otherwise to make good work.

When work is to be finished with two coats, the putty used for stopping the nail-heads and other indentations should be made of white lead, worked up with common whiting to the proper consistency, and the filling should be done after the first coat shall have become well dried. When more than two coats are to be applied, the filling should be done between the first and second coats, with ordinary pure linseed-oil putty.

It should be adopted as a rule, never to apply pure white as a priming coat; no matter whether the work is to be finished with one or four coats, the result will always be more satisfactory if the first coat be stained. A little finely-ground lampblack answers as well for this as anything.

The only way to produce solid, uniform work, is by making every succeeding coat lighter in tint than the one which preceded it. This is especially the case with walls, and other extended flat surfaces. No matter what the finish is to be, the first coat should always be darker than the one which succeeds it; and the darker the shade of the finishing coat, the more important it is that this rule should be observed. If the work is to be finished with black, prime with black. If with green, let that be the color of all the preceding coats. If with blue, let that color be the ground work. What can be more stupid than applying to work which is to be finished in imitation of black walnut a priming coat of white? All work should be primed especially with regard to the finishing color.

There is not half enough of dark colors used in priming applications. Venetian red, finely ground in boiled oil, deeply stained with black—and used very thin, in order to stain the wood as much as possible—is the best first coat for work which is to be finished in imitation of black walnut or other dark wood. The succeeding coats should be as dark as may be with a view to the proper shade of ground-work for the graining. In such case, if (as must happen in the ordinary course of events) the work becomes bruised or “chipped”—by an accidental knock from a chair leg or other article of house furniture—the general appearance of it is little impaired thereby. Quite the contrary, however, is the case if the underneath coats are white. Then, an accident of the kind before mentioned, shows a white spot, which staringly proclaims the work to be a delusion and a sham. Dark colors, too, as the Venetian red before mentioned, make better foundations than white lead or zinc. They dry harder and “rub” better, and, what is most important, cost less.

This matter having been duly considered, let us now proceed to the coats succeeding the first. Before applying a second coat, the first should be carefully rubbed, and all the nail-heads and other indentations carefully stopped with pure linseed-oil putty—using for flat surfaces a square-bladed putty-knife. Puttying with the fingers should never be tolerated (good work is now the subject under consideration). This done, the whole should be carefully examined to ascertain if the oil in the former coat shall have revealed any resinous or pitchy spots, not previously covered with the shellac. These preliminaries being attended to, the work may be considered ready for a second coat. The directions as to rubbing with sandpaper are to be observed in all the succeeding coats. As a rule, on interior work, paint should never be applied to a surface which has not been previously rubbed.

Sandpaper for fresh work and pumice-stone for old work. Always distrust the education of a painter in his trade who goes to work without a lump of pumice-stone, a sheet of sandpaper, a putty-knife, and a rag to wipe off the spatters—sparks, as the Irish not inaptly call them. Apropos of spatters! Every painter has seen (the result too of unpardonable negligence) plates of glass so covered with spatters, that to remove them would require more time than would serve to paint the woodwork of a “full-trimmed” window.

In priming work which is to be finished in oak, finely-ground French ochre is recommended. The objection to this pigment, that it does not work smoothly and easily under the brush, has arisen from its coarseness. Finely ground in boiled oil, it works as smoothly as white lead, and makes an excellent foundation for the succeeding coats.

For walls the first coat should be as dark in shade and as thin as practicable, the object being to stain the plaster as much as possible. Indeed, if the whole mass of plaster could be stained through and through, it would be desirable to so stain it.

The use of glue in wall painting is of doubtful propriety. It should never, under any circumstance, be put on until after the second coat, and then rubbed on with a rag, very lightly. In first-class work, however, its use is not recommended.

Plaster mixed with weak glue-size—which prevents its setting too rapidly—is the best material for stopping walls preparatory to painting, and each coat of paint should be carefully rubbed with worn sand-paper, before the succeeding coat is put on. For preparing walls a small pocket-trowel will be found a most serviceable tool, or a trowel-shaped putty-knife, which article has come into general use.

The preparation of ceilings for whitewashing (or kalsomining as this operation is sometimes pretentiously called) is an operation requiring some skill and knowledge of “how to do it.” A dirty ceiling, which has been subjected to successive coats of whitewash, whether of lime, or of whiting and glue-size, cannot be made solidly and smoothly white by additional whitewashing. The mass has become spongy, and sucks up the water so quickly that the material cannot be evenly distributed. In such case the only way is to begin anew, to go at once “down to hard pan” by removing all the previous applications by washing and scraping. This is best effected with a broad-bladed square-pointed putty-knife, keeping the ceiling wet meanwhile. Plaster (hard-finish) is not of uniform density, and some spots are much more absorbent than others. To remedy this a mixture of soft soap and alum, dissolved in water, should be applied with a broad kalsomine brush.

It is not assumed that mere verbal instructions can teach the art of whitening or tinting walls and ceilings in water-colors. To produce good results, great skill in preparing the materials and dexterity in manipulation are required; and such work should be intrusted only to competent hands. A mass of unsuitable material may be cheaply put upon a ceiling; but when the same shall require repainting, the cost of labor will be greater in removing the previous coating, than will be the whole cost of repainting. These remarks, too, apply equally to all kinds of painting; and reference is made to the whitening and tinting of ceilings only, because of the general impression that this kind of work may be performed by anybody.

The materials and tools used in painting are too costly to be wasted and worn by incompetent handling. "Painting just to keep the gardener or hostler out of idleness," will prove in most cases a left-handed economy. Such experiments are prudent only when the services of skilled workman cannot be obtained.

Taste in Color.—In rooms to be lived in, simple white for color of walls and paint, as well as any extremely dark treatment, should be avoided. The walls of rooms should be such backgrounds as will best suit the complexions and dresses of the larger number of people. Delicate white intensifies by contrast any unpleasantness or want of perfection; extreme dark would make people look white and ghastly. Neutral colors will be found the best—generally some grey or cool color that will contrast with warmth of complexions. On no account let an absolutely pure color be used for general surfaces. Nature provides no such color in pigments. Her yellows are greenish or reddish, and so on. Nor does she use it to any extent in inanimate nature. So much so that you will find that if you have much difficulty in describing a color, you may be certain it is good; the more difficulty the more beauty. Nature trusts mainly to gradations of tone, using vivid color in small quantities only, as in the touches on bright flowers and butterflies. This teaching of nature will be found seconded in the pictures of the greatest artists, and in following such teaching, it is necessary to consider the object to which (in domestic work, say) the rooms are to be devoted. A drawing-room, it is agreed, should be light, festive and gay; dining-room at once more sober, and with more depth and warmth, as befits its uses. You must also consider the light and shade; openings, and the positions of them; for these may (or may not) effect for you contrast of tone, and may even touch the question of the good sense of your whole scheme of decoration.

In a lecture delivered before the British Architectural Association, on this subject, the lecturer suggested that in the treatment of a drawing-room the walls should be a light neutral grey, fawn color, or pale green (not dark, but not white). Dados are suitable for all rooms, even drawing-rooms. They may be made of wood, painted as the room doors, or of stamped leather, or of the French paper imitations of stamped leather. A frieze does not interfere with the heads of sitters, and adds much interest if it has its sentiment or story. If flowers form part of your decorations, have no relief, no imitation of nature's light and shade. A wall must be a wall; if, neglecting this, you introduce illusions to the eye, the sense of solidity will not be suggested. The Japanese decorate on correct principles, with truth to the idea derived from nature, and truth in art, adaptation of representation to materials and method. As regards the woodwork there should be no graining anywhere; its aspect, however well executed, is repulsive. Real woods are always beautiful. Plain painting may be darker or lighter than the general wall surfaces; both will look well. The doors may have stencilled decorations in angles of panels; birds or butterflies, or plants, or any beautiful natural objects will supply motives. The ceilings should rarely be wholly white, except of halls or where the light is defective. Papered ceilings

look well. The use of gold is generally satisfactory; it reflects a warm tone on everything below. Put a good amount of color on a ceiling—not, however, making it so dark as to bring it too close to the eye. The carpet must be either lighter or darker than the walls. This is following out the artist's rule, to make either background or foreground run into the figure. If this is not done in painting, a woman in white satin, for instance, against a dark floor and dark walls, will look like a cut-out figure stuck on, and the same sort of result would occur in rooms. As in ordinary life dresses are dark in color, where a light wall tone has been recommended, the carpet will have to be darker than the walls. Not too vivid in color, however, and of course, no flowers, ferns, birds' nests, and such like fearful things. *Furniture and hangings* should not be too much alike in color; have, say, the carpet one tone, the coverings of the furniture another, and the curtains and other hangings a third. Have summer and winter hangings and furniture coverings; those for the former light and cheerful, the others with more warmth, and suggestive of comfort and home life. A table-cloth, occasional chair, or a rug, may supply a bit of effective contrast with prevailing hues of hangings, etc., and a spot of vivid color in a vase or some small hanging will complete the formal decoration of the room.

Graining.—The art of imitating the grain of the more expensive woods has been brought to a great degree of perfection, but of late years so many unskilled workmen have undertaken to imitate the natural grain of wood with such imperfect results, that this beautiful branch of painting has fallen into partial disuse. A few remarks, therefore, to the uninitiated may not be inappropriate in this work. Mahogany, satinwood, rosewood, mottled and walnut roots, maple, and some others, are frequently imitated; and it is seldom that a house is finished without some graining being introduced. The imitation of the above-mentioned woods are best performed in ground distemper (water colors) which are always preferable and more economical purchased ready prepared. Oak, chestnut, ash and similar long-grained woods, are best executed in oil-color, particularly for outdoor work. For drawing rooms, delicate party colors are preferable, as they harmonize better with the neutral tints on the walls or delicate tinted papers. The process of graining is very simple. To be an accomplished grainer, practice and an artistic taste is very necessary. Too frequently the imitation is overdone, the shading too deep and obtrusive, and the work made too glaring with figure and varnish, has an unnatural appearance; a grainer should always avoid attempting to over-do nature. The following is an approved method:—The surface on new wood should be prepared with three coats of oil paints for the ground color, and regulated in shade by the color of the wood to be imitated, making due allowance for the graining tint that is to cover it. The ground colors should always be perfectly dry before the graining is commenced. The painter then preparing small quantities of the colors he requires, applies it thinly and evenly over the surface and proceeds to wipe out with his thumb and a piece of white cotton cloth the figure of the grain. Some grainers use rubber instead. We cannot here explain all the different processes for the imitation of the grain of wood. Many painters have a method of their own, which from long practice produce excellent results. In some cases, graining in distemper may be adopted with great success for indoor work, and if the colors are put on thin, so that the varnish will penetrate through into the ground color, this kind of graining is as durable as oil-color, and is susceptible of being made far more beautiful, and soft looking in the imitation of mottled woods. For blending distemper colors, a badger's hair blender should be frequently used to soften down and blend the tints where necessary; but for blending oil-color, in order to

produce an elongation of the grain, we have found a flat varnish brush, kept moderately damp and clean, preferable to the badger's hair. When the work is dry, the shades necessary for some woods should be laid on in distemper-color (some use thin oil-colors) and then covered with two coats of good oil varnish. Common varnish should never be used on outside work.

Oak Graining.—In oak graining the color is made in the following manner:—Procure some finely-ground burnt umber and raw sienna (or Vandyke brown and raw sienna if a dark oak be required), and thin with equal parts of linseed oil and turpentine. Add a large quantity of patent dryer to make it stand the comb. The color is now ready for use. The graining color is brushed over the work in the ordinary way with a pound brush, care being taken not to put too much color on, else it is liable to look dirty. A dry dusting brush is now used to stipple with, which, if properly done, will distribute the color evenly. It is now ready for combing. First take a medium or coarse-cut gutta-percha comb, and draw it down one side of the panel, then use a finer one to complete it. This comb will leave the marks of the grain in clear unbroken lines from top to bottom of the panel. We now take a fine steel comb and go over the whole of the previous combing; but in drawing this comb down, we either move it in a slanting or diagonal direction across the previous combing, or draw it down with a quick and short wavy motion and curl. Both the former and latter motions will break up the long lines left by the gutta-percha comb into short bits, which, of course, represent the pores or grains of the real wood. Next take out the lights of figuring or veining. This is effected by means of a piece of washleather, held tightly over the thumb nail. Every time a few lights are wiped out the leather should be moved slightly, so that the same part of the leather will not be used twice, thus ensuring clean work. There are various methods of doing this, but they require much more practice. When the figures are all wiped out they will require to be softened. By softening we mean the imitation of those half shades seen upon and about the figures in the real wood. These are imitated by doubling a piece of washleather into a small roll, and with the side of this the grain is partly wiped away or softened. Care should be taken not to wipe off the whole of the grain. If the operator has a piece of the real wood to look at occasionally he will be materially assisted. As soon as the oil color is dry it should be over-grained. This is effected in water color. Next go over the work with a bit of sponge and soap to prevent it "cissing." Before laying on the over-graining, wash out the sponge and wipe the work. It is now ready to receive the color. Grind up finely a little vandyke brown in water, and dilute it with table-beer and water. It is then ready. Take a flat hog-hair brush, 3 in. to 4 in. wide, dip it in the color and draw it over the work, in most cases in the direction of the combing, but occasionally crossing. The hair of the brush, being thinly placed, will separate into patches, and hence the color will be deposited in streaks, resembling the natural gradations which the wood presents. If you have not a brush of this kind a sponge may be used to put in the streak and to soften off. Then dry varnish in the usual way.

Spirit Graining for Oak.—2 lbs. whiting, $\frac{1}{4}$ lb. gold size, thinned down with spirits of turpentine, then tinge your whiting with vandyke brown and raw sienna ground fine. Strike out your light with a pitch or piece of rubber dipped in turpentine, tinged with a little color to show the lights. If your lights do not appear clear, add a little more turpentine. Turpentine varnish is a good substitute for the above mentioned. This kind of graining must be brushed over with beer with a clean brush before varnishing. Strong beer must be used for glazing up top graining and shading.

Old Oak in Distemper.—To make an exceedingly rich color for the imitation of old oak, the ground is a composition of stone ochre or orange chrome and burnt sienna; the graining is burnt umber or vandyke brown, to darken it a little. The above colors may be used for oil as well.

Pollard Oak.—Ground color, a mixture of chrome yellow, vermilion and white lead, to bring it to a rich light buff. The graining colors are vandyke brown and small portions of raw and burnt sienna and lake, ground in beer or vinegar. Fill a large brush with color and spread it over the surface to be grained, and soften with a badger hair brush. Take a moistened sponge and dapple round and round in kind of knobs, then soften very lightly, after which draw a softener from one set of knobs to the other while wet, to form a multiplicity of grain, and finish the knobs with a hair pencil, in some places in thicker clusters than others. When dry, put the top grain on in a variety of directions, and varnish with turpentine and gold size; then glaze up with vandyke and strong beer. Finish with copal varnish. This is for distemper only.

Mottled Mahogany.—The ground is prepared with the best Venetian red, red lead, and a small proportion of white lead. The graining colors are burnt sienna, ground in beer, with a small portion of vandyke brown. Cover the surface to be grained, soften with a badger hair brush, and while wet take a damp sponge and go over the lights a second time, in order to give a variety of shade; then blend the whole of the work with the badger softener. Put the top grain on with the same color. When dry, varnish. For distemper only.

Rosewood.—Mix vermilion and a small quantity of white lead for the ground. Take rose-pink, tinged with a little lamp-black or vandyke brown, and grind very fine in oil, then take a flat graining brush, with the hairs cut away at unequal distances, and cut down the grain as if wending round a knob. When nearly dry, take a graining comb that is used for oak, and draw down the grain. This will give it the appearance of nature. Then varnish. This makes an excellent and durable imitation.

Another for Rosewood.—This ground is prepared with vermilion and small quantities of white lead and crimson lake. When the ground is dry, and made very smooth, take vandyke brown, ground in oil, and with a very soft tool spread the color over the surface in different directions, forming a kind of knots. Before the work is dry take a piece of leather, and with great freedom strike out the light veins; having previously prepared the darkest tint of vandyke brown or gum asphaltum, immediately take the flat graining brush with few hairs in it, called a top grainer, and draw the grain over the work and soften. When varnished, the imitation will be excellent.

Curled Maple in Oil.—Prepare a rich ground by mixing chrome yellow, white lead and burnt sienna. For the graining color, grind equal parts of raw sienna and umber with a little burnt copperas and turpentine, and mix it with a small quantity of grainer's cream, thin the color with oil; then fill a tool and spread the surface even and rub out the lights with sharp edge of a piece of buff leather, wiping it frequently to keep it clean; soften the edges of the work very lightly, and when dry, put on the top grain with burnt umber and raw sienna ground in beer with the white of an egg beat into it. Varnish.

Curled Maple in Distemper.—Prepare a light yellow for the ground, by mixing chrome yellow and white lead, tinged with Venetian red. The graining color is a mixture of equal portions of raw sienna and vandyke brown, ground in beer. Spread the surface to be grained in

an even manner; then with a piece of cork rub across the work to and fro, to form the grains which run across the wood; soften, and when dry lightly top grain with the same color. Varnish.

Bird's Eye Maple in Oil.—The ground is a light buff, prepared with white lead, chrome yellow and a little vermilion or English Venetian, to take off the rawness of the yellow. The graining color is equal parts of raw umber and sienna, ground in oil to the proper consistency. Spread the surface of the work with this color, and having some of the same prepared a little thicker, immediately take a sash tool or sponge and put on the dark shades, and soften with a badger hair brush; before the color is dry put on the eyes by dabbing the dotting machine on the work, or by striking the colors short and sharp with the tips of the fingers, then blend slightly the eyes in one direction only. When dry, put on the grain with the camels hair pencil on the prominent parts to imitate the small hearts of the wood. The same graining colors to be ground in here for distemper.

Walnut in Oil.—The ground is formed with ochre, Indian red, umber, and white. The graining coat is similar to that described under the oak heading, and is prepared with vandyke brown; and for the darker shades, fine ivory-black; the wiping out and blending to resemble that in mahogany; the fine dark veins of ivory-black to be lightly and wavily drawn over the work after it is blended. It is then ready for the varnish.

Satin-wood in Distemper.—This ground is prepared with white lead, stone ochre and small quantities of chrome yellow and burnt sienna. The graining color is one-third of raw sienna and whiting, ground in pale ale, very thin; then spread the color over the surface to be grained. While wet soften, and have ready a wet roller or mottling brush, in order to take out the lights; blend the whole with the badger hair brush, and with the same color put on the top grain. Varnish.

To Imitate Granite in Oil.—For the ground color, stain your white lead to a light lead color, with lamp-black and a little rose-pink. Throw on black spots with a coarse brush or graniting machine. A pale red also, and fill up with white before the ground is dry.

Another for the Same.—A black ground; when half dry throw in vermilion, a deep yellow, and white spots.

Marble.—For *White Marble*, get up a pure white ground, then hold a lighted tallow candle near the surface, and allow the smoke to form the shades and various tints desired. This will make a very handsome imitation. *Black Marble.*—Imitation is made by streaking a black surface with colors, using a feather and pencil. Another plan is to get up a smooth black surface; then take the colors, green, yellow, red, white, etc., ground thick in gold size, and streak the surface with a stick or hair pencil. Allow it to dry, and apply a heavy coat of lamp-black and yellow ochre mixed, mixed rough stuff. When all is hard, rub down to a level surface with lump pumice-stone, varnish and a beautiful variegated marble will be the result.

Red Marble.—For the ground put on a white tinged with lake or vermilion; then apply deep red patches, filling up the intermediate spaces with brown and white mixed in oil; then blend them together; if in quick drying colors, use about half turpentine and gold size. When dry, varnish, and while the varnish is wet, put in a multitude of fine white threads, crossing the whole work in all directions, as the wet varnish brings the pencil to a fine point.

Jasper Marble.—Put on a white ground lightly tinged with blue; then put on patches of rich reds or rose-pink, leaving spaces of the white ground; then partly cover these spaces with various browns to form fossils, in places running veins; then put in a few spots of white in the centre of some of the red patches, and leaving, in places, masses nearly white. When dry use the clearest varnish.

Blue and Gold Marble.—For the ground color put on a light blue; then take blue, with a small piece of white lead and some dark common blue, and dab on the ground in patches, leaving portions of the ground to shine between; then blend the edges together with a duster or a softener; afterwards draw on some white veins in every direction, leaving large open spaces to be filled up with a pale yellow or gold paint; finish with some fine white running threads, and a coat of varnish at last.

Black and Gold Marble.—This description of marble is very chaste, and is in great demand. The ground is a deep black, or a dead color, in gold size, drop black and turpentine; second coat, black japan. Commence veining; mix white and yellow ochre with a small quantity of vermilion to give a gold tinge; dip the pencil in this color, and dab on the ground with great freedom some large patches, from which small threads must be drawn in several directions.

In the deepest part of the black a white vein is sometimes seen running with a number of small veins attached to it, but care must be taken that these threads are connected with and run, in some degree, in the same direction with the thicker veins. If durability is not an object, and the work required in a short time, it may be executed very quickly in distemper colors, and when varnished, it will look well.

Compound Colors.—The following tints can be formed by mixing the colors as below. The shades can be made to suit any taste by the exercise of a little judgment in proportioning the colors:

Cream.—White lead, yellow and red.

Drab.—White, Prussian blue and vermilion.

Fawn.—White, stone ochre and vermilion.

Flesh.—Lake, white lead, and a little vermilion.

Grey, Pearl.—White lead, Prussian blue, and a very little black.

Grey, Flaxseed.—White lead, Prussian blue, and a little lake.

Gold.—Massicot, or Naples yellow, with a small quantity of Realgar and Spanish white.

Green, Light willow.—White, mixed with verdigris.

Green, Grass.—Yellow pink with verdigris.

Green, Pea.—White lead and chrome or Paris green.

Green, Dark.—Black and chrome green.

Green, Olive.—Prussian blue and French yellow; mix to the tints required. This is a cheap and handsome color for outside work, such as doors, carts, wagons, railway cars, etc.

Jonquil.—Yellow, pink and white lead. This color is only for distemper.

Lead.—Prussian blue and white, with a light shade of white.

Olive.—For distemper, use indigo and yellow pink mixed with whiting or white lead powder.

Olive.—Red, green, or black and yellow.

Pearl.—White lead, Prussian blue and red.

Purple.—Dark red mixed with violet.

Purple.—White, Prussian blue and vermilion.

Red, Dark.—English Venetian, red lead and litharge.

Red, Light.—Venetian red, and red lead in equal parts.

Red, Deep.—Vermilion, with a very small quantity of red lead.

Stone.—White, with a little spruce ochre.

Straw.—White lead and yellow.

Snuff.—Yellow, sienna and red.

Slate.—White lead, black, red and blue.

Steel.—Ceruse, Prussian blue, fine lac and vermilion.

Salmon.—White lead, yellow and red.

Walnut.—Tree color; two-thirds white lead and one-third red ochre, yellow ochre and umber, mixed according to the shade sought. If veining is required, use different shades of the same mixture. (See article on graining).

Yellow, Light.—French yellow and white lead. A little red lead may be used.

Another.—French yellow, white and red lead.

Another.—A mixture of a small portion of Prussian blue, French yellow, white lead and Turkey umber and burnt vitriol, or litharge, will produce different shades of yellow, according to the preponderance of one of the above colors.

Yellow.—Bright for floors, white lead, French yellow, chrome yellow a little, some red lead and litharge; mix with equal parts of boiled oil and turpentine and use it thin.

Yellow, Dark.—French yellow and a little red.

Yellow, Lemon.—Yellow pink, with Naples yellow. For distemper only.

MISCELLANEOUS RECEIPTS.

For Iron.—A good paint for preserving iron exposed to the weather, is made as follows:

Pulverized oxides of iron, such as yellow and red iron ochres, or brown hematite iron ores, finely ground, and simply mixed with linseed oil and a dryer.

White lead applied directly to iron is thought to have a corrosive effect. It may be applied over more durable colors.

Red lead, when pure, is very durable. An instance is recorded of iron painted with it having been under water for nearly 50 years, and had not been affected by rust.

Sheet iron, before being used for roofs or other outside purposes, should be heated and dipped into hot linseed oil, which will penetrate into it. Tinned iron in roofs has been found to corrode quicker than in former years, owing to the more general use of coal.

Paint for Rusty Iron.—Black Japan varnish, mixed with turpentine, to make it thinner if necessary, is one of the best preventatives; but the iron must be dry when you put it on. If you can warm the iron when painting it, so much the better. If not sufficiently opaque, you may put in dry finely pulverized paint, such as lamp-black. Red lead, with linseed oil is also a good paint for rusted iron; so are the mineral reddish-browns which consist of oxide of iron; they become very hard, and are used for the iron-work of the elevated railroads in this city.

To Paint on Stucco.—Great care is required in painting upon stucco, for the work must not only be thoroughly dry, but free from any liability of dampness; that is to say, the walls themselves must be dry. It is, consequently, usual to allow the stucco to remain for several months before it is painted; and this is especially necessary when it covers over a large surface, as in the walls of churches, chapels and theatres. If the paint be applied too soon, the work will have a blotched appearance, and be

probably filled with small vesicles, formed during the evaporation of the water. When the work is dry, it may be prepared by covering it with a coat of linseed oil, boiled with dryer. This must be laid on very carefully, or the face will be irregular. The color may then be applied, and four coats will not be too much, the work being new. Persons are generally so anxious to have their buildings finished, that they disregard the future appearance of the work, and within a few weeks after the application of the stucco, cover it with paint. But it would, in all cases, be sufficient to wash the surface with distemper, as it would give a finished appearance to the building, and make it less necessary to hurry the work. When the work is sufficiently dry to receive the oil-colors the distemper color should be removed by washing, and when the stucco is dry apply the oil-color. The tints may be regulated by mingling different colors, as in all other kinds of painting.

Plastered walls should not be painted until they are thoroughly dry, and all settling in a new house has taken place. If painted too soon they will blister.

Unseasoned wood should never be painted, as it stops the pores of the wood and the sap acidulates, causing dry rot. Greasy surfaces must be washed with water mixed with lime or soda, otherwise the paint will not adhere to it.

Mixing Quick Drying Paint.—Venetian blinds should be painted to dry dead, then varnish; but few take this trouble. Mix the paint as under: White lead, boiled oil, and the least drop of turps; mix sufficient of each to form a creamy mixture; then add about 1 oz. patent drier to each 1 lb. of paint. If you want the paint darker use enough burnt umber to give the required tint. If you want to varnish, omit the oil and use turps.

Transparent Paint for Glass.—Take for blue pigment, Prussian blue; for red, crimson lake; for yellow, Indian yellow; and for other shades, a mixture of the appropriate primary colors. Rub them in a size made as follows: Venice turpentine, 2 parts; spirits of turpentine, 1 part, and apply with a brush. The colors are moderately fast unless exposed too long to direct sunlight. A solution of the various aniline dyes in shellac varnish has also been recommended.

Gilding.—Gold leaf is the only successful application. First put on a coat of Japan gold size, and when that is “tacky,” and nearly dry, lay on the gold-

leaf and dab it with a small tuft of cotton-wool. Where you buy the gold-leaf you can buy a gilder's tool for applying it; but in order to get a smooth surface it must (when perfectly dry) be burnished with an agate burnisher, which you will also get at the color-shop, but you will not be able to burnish gold-leaf on the bare wood. You can, if you like, varnish with pale copal varnish.

Gilding on Glass.—Glass letters are gilt the same way as you would a name on a glass door. You can easily get a good burnish if you take a little trouble. Get some of the best cotton wool at a chemist's, and well polish the gold with it; the gold must be thoroughly dry. Then go over it with your size boiling hot; do not touch the same place twice with the brush, or you will bring the gold up; repeat the process three or four times, being sure to have your gold dry each time, the hotter the size the brighter will be the burnish; be careful, however, and not break the glass with the heat.

Gilding Fretwork, etc.—The first thing to be done is to whiten the work. To do this scrape some whitening very fine, place it in a pipkin with a lump of gilder's size, and water sufficient to make it of the consistency of thick cream, when heated over a fire; then, with a camel-hair pencil, paint it on the object several times, allowing each coat to dry before applying the next. When the several coatings have raised it to the thickness of $\frac{1}{16}$ in., set it aside for twelve or more hours, to harden; when hardened, smooth the surface with very fine sandpaper first, and finally with a piece of cork; when using the cork frequently dip it in water, and, when practicable, use it in a circular motion. Thus far successful, the next thing is to lay on the gold. To gild, then, dissolve some gilder's—not common size—in water, and heat, and with a full brush lay it on the surface of the object. Cut the gold leaf, on a pad of buff leather, with a clean cut of the knife (much easier said than done; perseverance, however, with the cost of a book or two of gold mutilated, and a large amount of patience exhausted, will overcome the difficulty), to the size required; take these up on a tip (a row of long hairs placed between two bits of cardboard)—the professional way to do this is to strike the hair of the tip against the gilder's own whiskers or hair—and gently lay them on the surface of the object, taking care that each succeeding piece slightly overlaps the preceding. When dry, a small piece of fine sponge, dipped in a weak solution of size water, should be gently passed over it to give a uniform appearance. If the bright gold requires to be

deadened, deep ormolu should be used in a similar way after sizing. The yellow used for the ungilt portions consists of gilder's yellow, dissolved in size water, and is put on with a brush.

Painting on Gilded Panels.—There is no preparation needed to paint in oils on a gilded panel. No mediums are required, the ordinary oil colors being used unmixed. If required to dry flat and to remain so, they are mixed with turpentine and left unvarnished. If the shiny look of oils is to be retained, they are slightly diluted with boiled oil, and varnish with white hard varnish when dry.

Gilding on Wood.—To gild in oil, the wood, after being properly smoothed, is covered with a coat of *gold size*, made of drying linseed oil mixed with yellow ochre; when this has become so dry as to adhere to the fingers without soiling them, the gold leaf is laid on with great care and dexterity, and pressed down with cotton wool; places that have been missed are covered with small pieces of gold leaf, and when the whole is dry, the ragged bits are rubbed off with cotton. This is by far the easiest mode of gilding; any other metallic leaves may be applied in a similar manner. *Pale leaf gold* has a greenish yellow color, and is an alloy of gold with silver. Dutch gold leaf is only copper colored with the fumes of zinc; being much cheaper than gold leaf, is very useful when large quantities of gilding are required in places where it can be defended by the weather, as it changes color if exposed to moisture, and it should be covered with varnish. *Silver leaf* is prepared every way the same as gold leaf; but when applied, should be kept well with varnish, otherwise it is liable to tarnish; a transparent yellow varnish will give it the appearance of gold.

Whenever gold is fixed by means of linseed oil, it will bear washing off, which burnished gold will not.

To Gild Letters.—When the sign is prepared as smooth as possible, go over it with a sizing made by white of an egg dissolved in about four times its weight of cold water; adding a small quantity of fuller's earth, this to prevent the gold sticking to any part but letters. When dry, set out the letters and commence writing, laying on the size as thinly as possible, with a sable pencil. Let it stand until you can hardly feel a slight stickiness, then go to work with your gold leaf knife and cushion, and gild the letters. Take a leaf upon the point of your knife, after giving it a slight puff into the back part

of your cushion, and spread it on the front part of it as straight as possible, give it another slight puff with your mouth to flatten it out. Now cut it to the proper size, cutting with the heel of your knife forwards. Now rub the tip of the knife lightly on your hair; take up the gold on the point, and place it neatly on the letters; when they are all covered, get some very fine cotton wool, and gently rub the gold until it is smooth and bright. Then wash the sign with clean water to take off the egg size.

Sign Writing in Colors, Etc.—On an oak ground ornamental letters, in ultra-marine blue, filled in with gold and silver leaf, blocked up and shaded with burnt sienna. *Another.*—Gold letters on a white marble ground, blocked up and shaded with a transparent brown or burnt sienna. *On glass.*—Gold letters shaded with burnt sienna. *Another.*—Gold letters shaded with black on a *scarlet* or *chocolate ground*. On a rich *blue ground* shaded with black, look very well. *On a purple ground*, pink letters shaded with white. Mix ultra-marine and vermilion for a ground color, white letters shaded with grey. *Vermilion ground*, chrome yellow stained and vermilion and lake, for the letters shaded with black.

A substitute for the above colors: Rose-pink and red lead; and for the letters stone yellow, white lead and Venetian red. Mix your colors for writing in boiled oil, and use for dryer gold size. Other good grounds for gold letters are, blues, vermilion, lake and Saxon. When your sign is ready for gilding, follow the directions under the head “*To Gild Letters on Wood.*”

Gilder’s Size.—Drying or boiled linseed oil, thickened with yellow ochre, or calcined red ochre, and carefully reduced to the utmost smoothness by grinding. It is thinned with oil of turpentine.

Staining Wood a Dull Black.—The work required to be stained should be colored with drop-black and size. When this is thoroughly set it should be papered off and colored again, and then be papered off again. The polish should also be stained with drop black and a little indigo. Next polish to a perfect surface, and let it set. After the wood has absorbed all the polish possible, polish again, and dull it with the finger dipped in fine emery; a fine metallic surface will thus be obtained.

Staining Floors.—The best and cheapest and only permanent stain for floors is permanganate of potash. Buy it by the $\frac{1}{4}$ lb., and at a wholesale chemist’s; mix about $\frac{1}{4}$ oz. in a quart of water. Apply freely and quickly to

a dry floor with either cloth or brush, the latter if you care for staining your hand. Repeat the process for a very dark oak color; when dry, oil with burnt oil or beeswax and turpentine; you cannot wash this color out. Benson's stain is only permanganate of potash. At first for a few moments the color is bright magenta, but this at once changes to a dark permanent brown. For fifty cents a whole house may be stained.

Varnishing Wood.—After smoothing wood with veneer scraper, brush on thick coat of shellac varnish; then use fine sandpaper, No. 0. Do this three times for close grained woods, such as black cherry, and four times for porous wood, such as chestnut. Have two dishes. Into one put finely ground pumice; into the other raw or boiled oil. Apply a mixture of these with a piece of hair-cloth or broad-cloth. Don't rub too hard. Finish up with rotten stone, which will remove pumice and oil. Above is a good dead varnish. *Another.*—Take encaustic wax, heat, and apply with a cork; rub in well, brush on thin coat shellac varnish, finish with pumice and oil.

Solvent for Old Putty and Paint.—Soft soap mixed with a solution of potash or caustic soda, or pearlash and slaked lime mixed with sufficient water to form a paste. Either of these laid on with an old brush or rag, and left for some hours, will render the putty or paint easily removable. *Another.*—Slack three pounds of stone quicklime in water, then add one pound of pearlash, and make the whole about the consistence of paint. Apply it to both sides of the glass, and let it remain for twelve hours, when the putty will be so softened that the glass may be easily taken out of the frame. *Another.*—Break the putty up in lumps the size of a hen's egg, add a small portion of raw linseed oil, and water sufficient to cover the putty, boil this in an iron vessel for about ten minutes and stir it when hot. The oil will mix with the putty, then pour the water off and it will be like fresh made. For removing hard putty from a window sash take a piece of square iron, make the same red hot, and run it along the putty till it gets soft. The putty will peel off without injuring the wood work.

Wash for Outside Work.—For woodwork slake half a bushel of fresh lime, by pouring over it boiling water sufficient to cover it 4 or 5 inches deep, stirring it until slacked; add 2 lbs. of sulphate of zinc (white vitrol) dissolved in water. Add water enough to bring all to the consistency of thick whitewash; it may be colored by adding powdered ether, Indian red, umber, etc. If lampblack is added to colors, it should first be thoroughly dissolved

in alcohol. The sulphate of zinc causes the wash to become hard in a few weeks.

Another for Brick, Masonry, and Rough-cast. Slake half a bushel of lime as before; then fill a barrel $\frac{2}{3}$ full of water, and add a bushel of hydraulic cement. Add 3 lbs. of sulphate of zinc previously dissolved in water. The whole should be of the thickness of paint. The wash is improved by stirring in a peck of white sand, just before using it. It can be colored as before described.

French Polish.—Coat with one or more coats of shellac and rub down smooth; make a rubber by rolling up a piece of flannel about 3 or 4 inches wide until it is about $1\frac{1}{2}$ inches in diameter, and tie it round with cord. Lay the end of the rubber on the mouth of a thin necked bottle and apply the varnish to it, having previously shaken up the contents in the bottle; then enclose the end of the rubber with a piece of soft linen doubled, and moisten the face of the linen with a little raw linseed oil.

Pass the rubber with a quick, light and circular motion over the surface until the varnish becomes dry, or nearly so, and charge the rubber again with varnish until 3 coats have been laid on, when a little oil may be applied to the rubber and two more coats laid on. In the finishing coat wet the inside of the cloth with a little alcohol, and rub quickly and lightly over the whole surface. Lastly wet the linen cloth with a little oil and alcohol without varnish, and rub as before until dry.

The varnish is the usual preparation of shellac. See cabinet maker's varnish.

Wood Filling Composition.—Boiled linseed oil, 1 qt.; turpentine, 3 qts.; corn starch, 5 lbs.; Japan, 1 qt.; calcined magnesia, 2 oz.; mix thoroughly. *Another.*—Whitening, 6 oz.; Japan, $\frac{1}{2}$ pt.; boiled linseed oil, $\frac{1}{2}$ pt.; turpentine, $\frac{1}{2}$ pt.; corn starch, 1 oz.; mix well together and apply to the wood. Add coloring if required. *Another.*—Linseed oil, 1 qt.; spirits of turpentine, $\frac{1}{2}$ pt.; lime, the size of a base-ball, broken fine. Let the mixture simmer on a stove, covered over, for two or three hours, then strain through a coarse cloth. It is to remain on 24 hours, then rub off with a woolen cloth and polish.

German Filling.—Fill the pores with raw tallow and plaster of Paris well amalgamated before a fire in cold weather. Darken, if required, with any coloring to suit. When well rubbed in give a coat of shellac and French polish or varnish.

Polish for Walnut Wood.—Mix with two parts of good alcoholic shellac varnish, one part of boiled linseed oil, shake well, and apply with a pad formed of woolen cloth. Rub the furniture briskly with a little of the mixture until the polish appears.

Rules the Painter Should Observe.—Never eat or sleep without washing the hands and face and rinsing the mouth. Keep the buckets, brushes, etc., clean, so that they may be handled without smearing the hands. Never sleep in a paint shop nor in a newly painted room. Never allow paint to accumulate on the clothing or finger nails. Never wash the hands in turpentine, as it relaxes the muscles and injures the joints; any animal oil or even linseed oil is better. Never drink water that has stood any length of time in a paint shop or newly painted room. Never use spirituous liquors as it unites with the mineral salts and tends to harden them and causes inflammation of the parts where they concrete. Milk, sweet oil and the like should be used freely, as they tend to soften the accumulated poisons and carry them off. Vinegar and acid fruits used constantly, unite.

PAPER-HANGING.

The art of putting on, or “Hanging” paper is very simple, and is easily learned; but to make a tasteful choice of paper for various situations, is not so easy, hence the following remarks, which may be of service to the workman or others on whom the selection of paper may devolve.

Walls to a room should be regarded only in the light of a frame-work to what the room contains, and should be decorated so as to bring into prominence and not eclipse the other parts of the chamber. Nothing destroys the effect of a room so much as a handsome but staring wall paper, or a wall so profusely ornamented as to strike upon the eye to the exclusion of the rest of the decorations, thus bringing forward what should be the background into the most conspicuous place. A modern drawing room is always difficult to decorate artistically, because of the taste of its builders for heavy cornices, prominent mantelpieces, and rooms too lofty for their size; and as all these misnamed “embellishments” are too costly to remove by tenants, the only plan to pursue is to destroy their effect by exercising both taste and ingenuity. First, with regard to the ceiling, the ornamental plaster boss in its center should be removed, and the ceiling tinted a color that harmonizes with the wall paper, as no harmonies can be hoped for when what produces them is surmounted with the glaring white of an ordinary ceiling. The tint used must be one that softens into the wall paper, not one that contrasts; thus, if the tone of the room is that of a soft grey blue, the ceiling should be a clear flesh pink; or should a grey gre picked out with black be the chosen color, then it should be colored a subdued lemon.

Some people cover their ceilings with a whole colored paper, and border it with a stencilled pattern representing the thin garlands so familiar upon Queen Anne decorations, but this is a more troublesome plan than the simple coloring, which answers all the purpose. The walls, if they are lofty, require a high dado. These high dados give a look of comfort and “home” that is absent from the modern high pitched room papered with one uniform

pattern. The dado is divided 3 feet to 4 feet from the ceiling, and the coloring of the lower portion must always be heavier than that used on the upper or a top heavy look will be given to the room. When many pictures are to be hung up the lower part of the dado should be of a whole color, either a whole colored paper or a painted wall, as pictures are only shown off upon such a background. Where a whole tint is used for the lower part of the dado, the upper portion should be decorated with a frieze paper of a good bold pattern, but of subdued coloring and of tint that harmonizes with the lower. Thus, the color used about the frieze should be the same as that on the lower part, but of a lighter shade, intermixed with some other colors that form a harmonious link between the two shades. Contrasts must be carefully avoided, but pale pinks, blue and ambers can be blended together above a subdued grey blue ground. The two portions of the dado should be joined together with a light wooden (black or brown) railing, or with a line of paint.

The dado decoration can be altered by placing the pattern paper upon the lower part and leaving the upper plain-colored with or without a stencilled pattern upon it. This will suit a room where not many pictures are required, or that is already rather dark. Some part of the wall should always be in plain color, as the eye requires rest; and no pattern, however subdued in hue, can give the relief to the mind that a bit of plain coloring affords, and this scarcity of ornament in one part of a room is amply repaid by the effect it gives to such parts as are bright and should be bright. The true theory of effect is to use but one or two bright colors in a room, and to surround them by soft and subdued tints that throw up and do not destroy their brilliancy; a number of bright colors placed together destroy each other, and leave no impression upon the mind but glare and vulgarity. Having settled upon your paper and ceiling, have the woodwork and cornice of the room painted either a shade lighter or darker than the walls, and shroud up the mantelpiece with curtains, etc., of satin sheeting embroidered with crewels, and instead of the usual looking glass over the fire-place, have a mirror surrounded with brackets holding china; or have a black wooden mantelpiece made with squares of looking glass set in. The back-ground of your room being thus completed in a manner really to be a back-ground, your furniture will look twice as well as if it were stared out of countenance by the walls, and one need hardly add that all your friends will delight in a

room that throws up and brings out their dresses and faces, instead of killing them by its glaring tints.

Operations.—To prepare the walls, make a size of glue and water, then give the walls a coat of a very weak solution of the same. To make a paste, take two pounds of fine flour, put in a pail; add cold water, and stir it up together in a thick paste. Take a piece of alum about the size of a small chestnut, pound it fine and throw it into the paste; mix well. Then provide about six quarts of boiling water and mix while hot with the paste until the whole is brought to a proper consistency. This makes an excellent paste, and fit for use when cold.

Cleaning Paper-Hangings.—A very good method to clean paper-hangings is that used oftentimes in cleaning the margins of prints. Cut into eight pieces a quartern loaf two days old, and after having swept all the dust from the paper-hangings, commence from the ceiling downward, and somewhat lightly rub the paper with a downward stroke with one of the pieces of bread. Continue this round the room, and then commence lower down in the same way till the whole of the surface of the paper has been gone over. The bread will from time to time get dirty, and it should be cut off as often as required. Care should be exercised not to rub the paper with a cross or horizontal stroke, only with the perpendicular movement of the hand or the paper may peel off the walls from the joints. With careful manipulation, paper will look almost as good as new.

Varnishing Paper-hangings.—When papers are hung in places where there is much passing, they are subject to greater wear than in an ordinary room, and varnishing is resorted to, to make them more durable, as it prevents soiling; they may also be cleaned with a damp cloth, and are not in wear so likely to be damaged by rubbing off the wall. Halls, lobbies, staircases, and offices are sometimes varnished for durability, but as a rule the appearance is not improved. The paper must be sized twice, the first being dry before the next is applied, when the varnish is laid evenly on. It is necessary to size the paper twice, as the first time may not effectually cover it, when the varnish would stain the paper as badly as if oil had been applied. The size necessary for sizing the paper before varnishing should be clean and transparent. Parchment cuttings boiled down in water and strained will make excellent size for this purpose. Before it is cold it should be either strained through calico, or poured off, leaving the bits at the bottom. A

patent size is sold for this purpose, but parchment size is the most reliable. Use clean brushes for size and varnish.

A Paste that will Keep Well.—A correspondent writes to a journal this formula for a non-poisonous paste that will keep well in hot weather: Wheat flour, 1 ounce; powdered alum, $\frac{1}{2}$ drachm; water, sufficient, or 8 ounces; oil of clove, or wintergreen, 3 or 4 drops. Rub the flour and the alum with the water to the consistency of milk; place this over a moderate fire, and stir constantly, until the paste drops from the wooden paddle in jelly-like flakes, and has the appearance of glycerite of starch. While the mass is still hot, add the essential oil, and pour the paste into an earthenware pot or open jar. In the course of about an hour a crust forms on the top; pour gently on this an inch of water, more or less. When some paste is wanted, decant the water, take out the quantity needed, and put some water again on the remainder, repeating the operation each time. Paste may be kept in this way for months, and will never be troubled with flies.

USEFUL HINTS FOR ESTIMATING COST OF WORK AND MATERIALS.

All surface painting is measured by the superficial yard, girting every part of the work covered, always making allowance for the deep cuttings in mouldings, carved work, railings or other work that is difficult to get at.

Where work is very high and scaffolding or ladders have to be employed, allowances must be made.

The following rules are generally adopted in this country in the measurement of work. Surfaces under six inches in width or girt are called 6 inches; from six to 12 inches, 12 inches; over 12 inches measured superficial. Openings are deducted, but all jambs, reveals or casings are measured girt.

Sashes are measured solid if more than two lights. Doors, shutters and paneling are measured by the girt, running the tape in all quirks, angles or corners. Sash doors measure solid. Glazing in both windows and doors is always extra. The tape should be run close in over the battens, on batten doors, and if the stuff is beaded, add one inch in width for each bead.

Venetian blinds are measured double. Dentels, brackets, medallions, ornamented iron work, balusters, lattice work, palings or turned work should all be measured double. Changing colors on base boards, panels, cornices, or other work, one-fourth extra measurement should be allowed for each tint.

Add five per cent. to regular price for knotting, puttying, cleaning and sand-papering.

For work done above the ground floor charge as follows: Add five per cent. for each story of 12 feet or less, if interior work; if exterior work, add one per cent. for each foot of height above the first 12 feet.

It is impossible to give prices for painting that will be reliable in any one section of this country for more than a month, as the cost of materials and

labor is continually varying, but we offer the following hints, which may assist the workman to make a fairly just estimate of work:

For plain colors on wood—

First coat, per yard,	\$—
2 coats, add "	4-5
3 coats, " "	1½

Now, by this rule, if the first coat was 10 cents, two coats would be 18 cents, and three coats would be 25 cents. This is a simple method and easily understood, and is adapted to any sort of plain painting or graining.

Graining, with one or two coats of good varnish, is worth, at this writing, from 65 cents to \$2.00 per yard, and marbling from \$1.50 to \$2.75 per yard.

Marbling mantels is worth from \$5.00 to \$20.00, according to style of marble and beauty of finish. All of the above prices are based on the understanding that the workman furnishes all tools and materials.

Transcriber's Notes

Missing and incorrect punctuation has been fixed.

[Page 8](#): “in cold whether” changed to “in cold weather”

[Page 29](#): The 2 sentences beginning with “The materials and tools used” are missing words in this and later editions and have been left as per the original.

[Page 52](#): “wollen cloth” changed to “woolen cloth”

[Page 54](#): The sentence beginning with “The tint used” has an incomplete word in this and later editions and has been left as per the original.

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